CS & IT ENGINEERING

DIGITAL LOGIC

MINIMIZATION



Lecture No. 5



By - CHANDAN SIR

Recap of Previous Lecture



K-MAP

Topics to be Covered



PRIME IMPLICANTS AND ESSNTIAL PRIME IMPLICANTS

- PRIME IMPLICANTS
- ESSENTIAL PRIME IMPLICANTS
- Questions Practice
- Discussion

Don't care :→

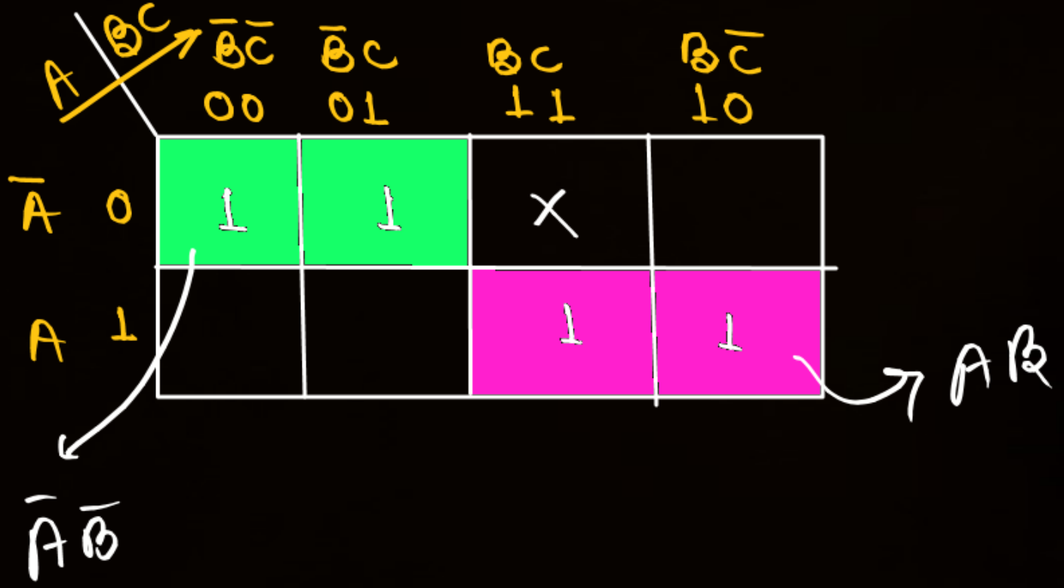
↳ combination of inputs on which o/p may or may not depends.

$$f(A.B) = \bar{A}\bar{B} + \bar{A}B + (AB)$$

Ex $f(A,B,C) = \Sigma m(0,1,6,7) + \Sigma d(3)$

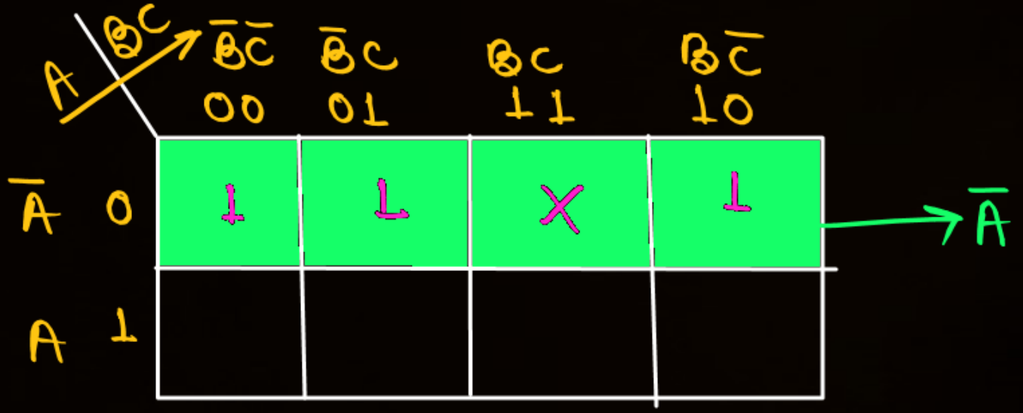
CJ SIR

Birthday = 31st oct



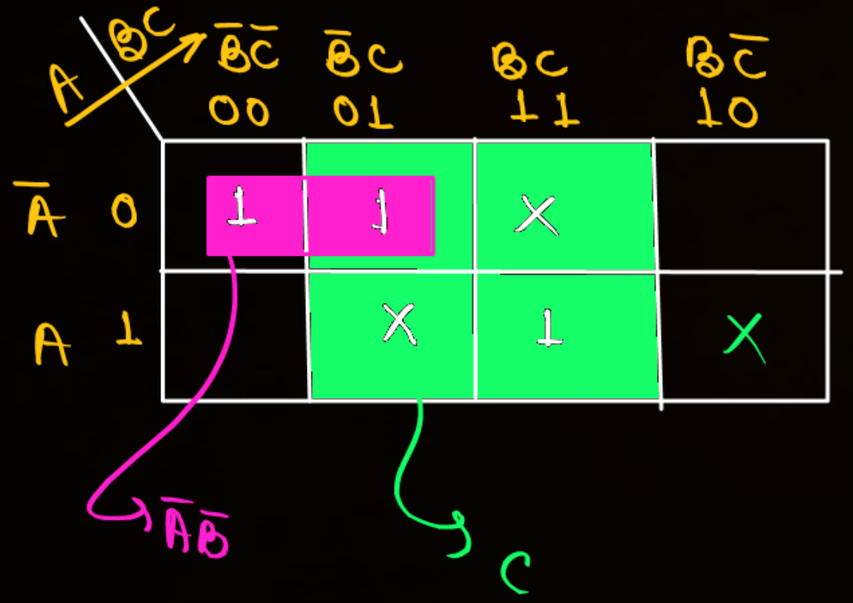
$\bar{A}\bar{B} + AB$

Ex $f(A,B,C) = \Sigma m(0,1,2) + \Sigma d(3)$



$= \bar{A}$

Ex $f(A,B,C) = \Sigma m(0,1,7) + \Sigma d(3,5,6)$

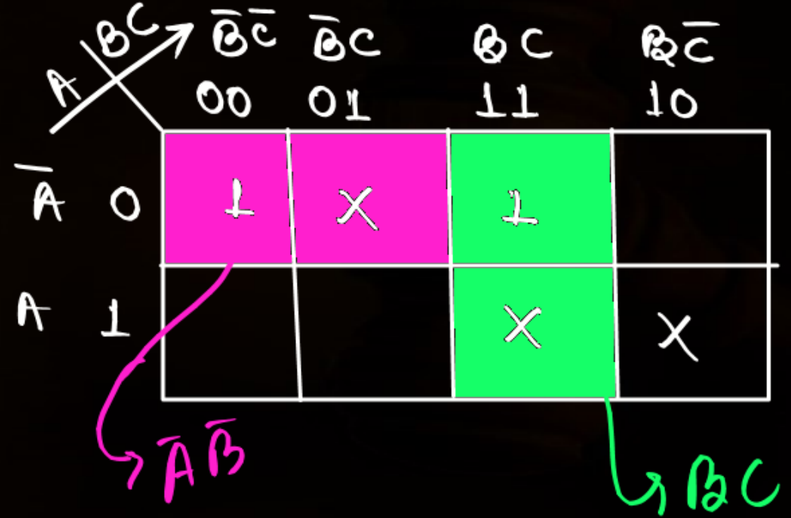


$= \bar{A}\bar{B} + C$

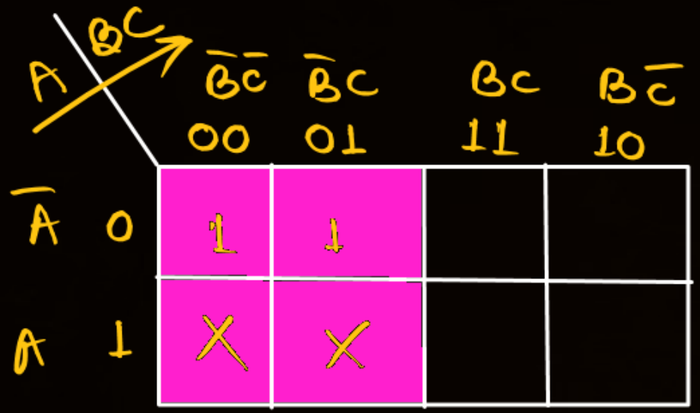
Question :→

A \ BC	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	1	X	1	
A 1			X	X

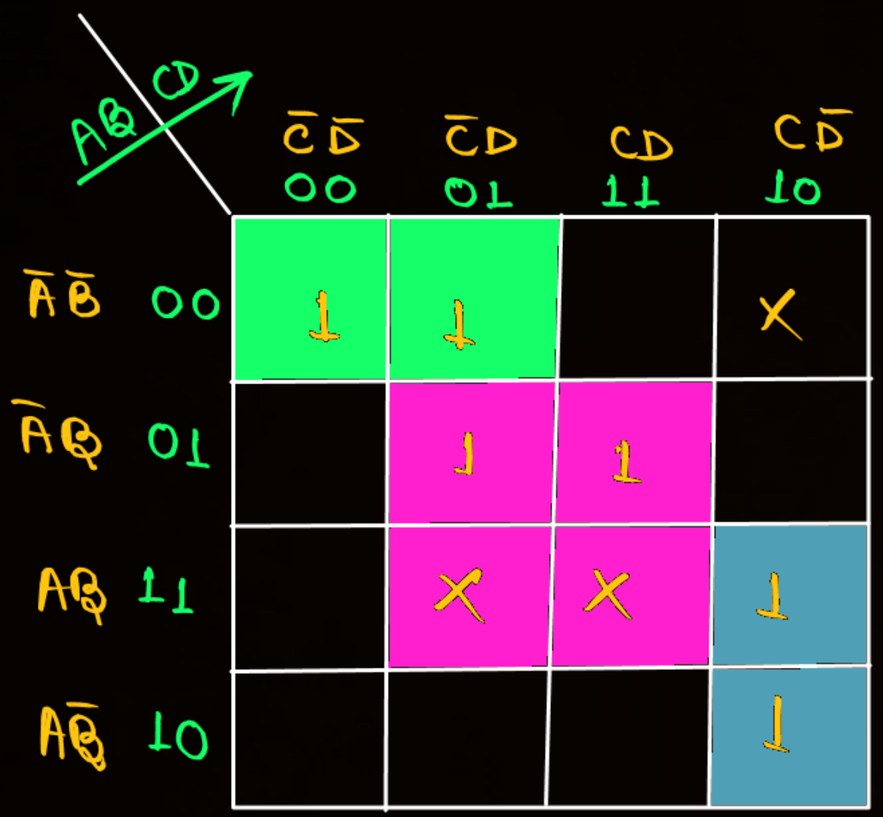
$= \bar{A}\bar{B} + \bar{A}C$



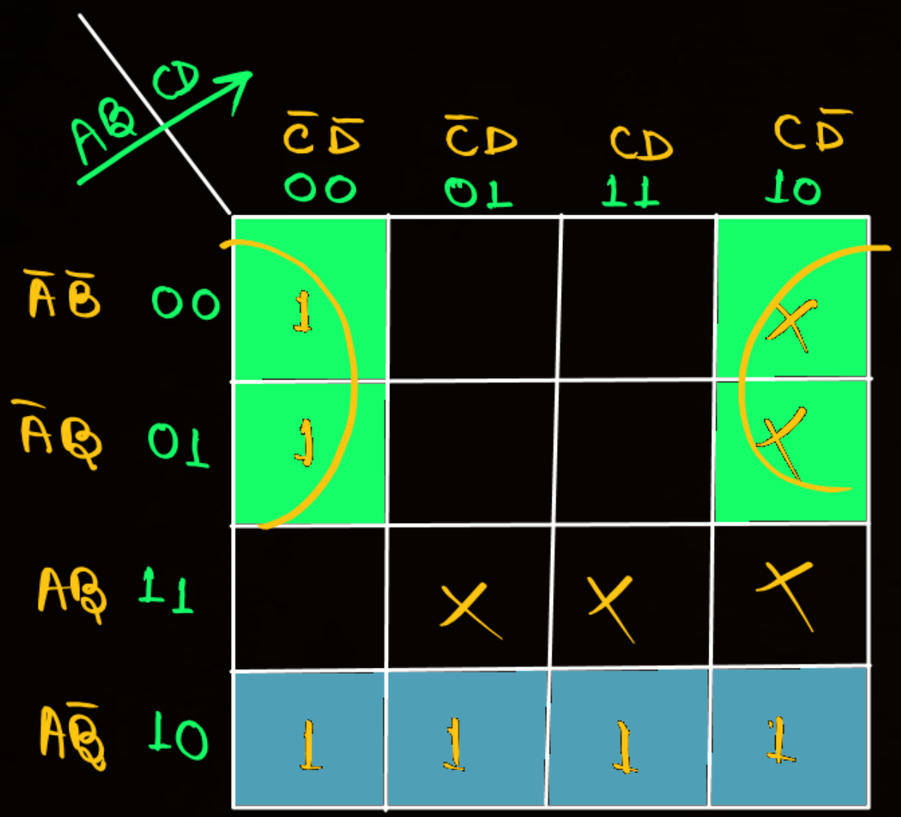
$= \bar{A}\bar{B} + BC$



$= \bar{B}$

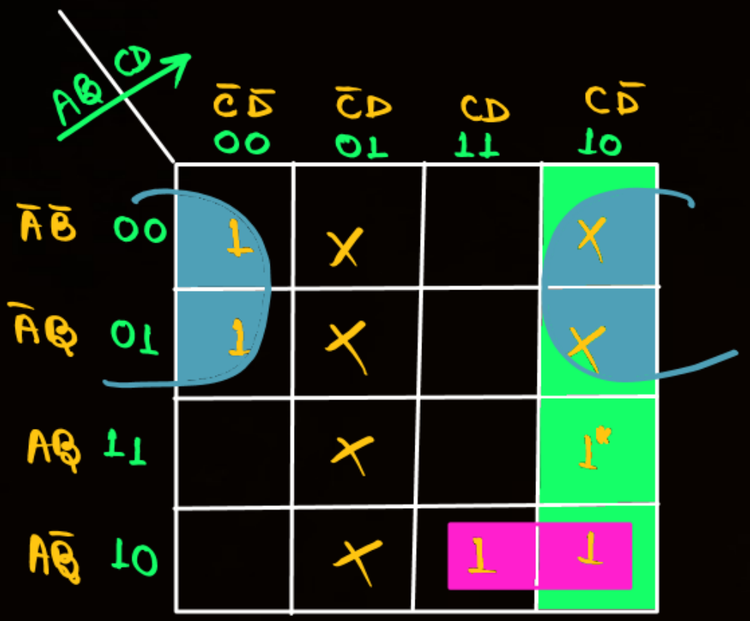


$$\bar{A}\bar{B}\bar{C} + BD + AC\bar{D}$$



$\bar{A}\bar{D} + A\bar{B}$

AB

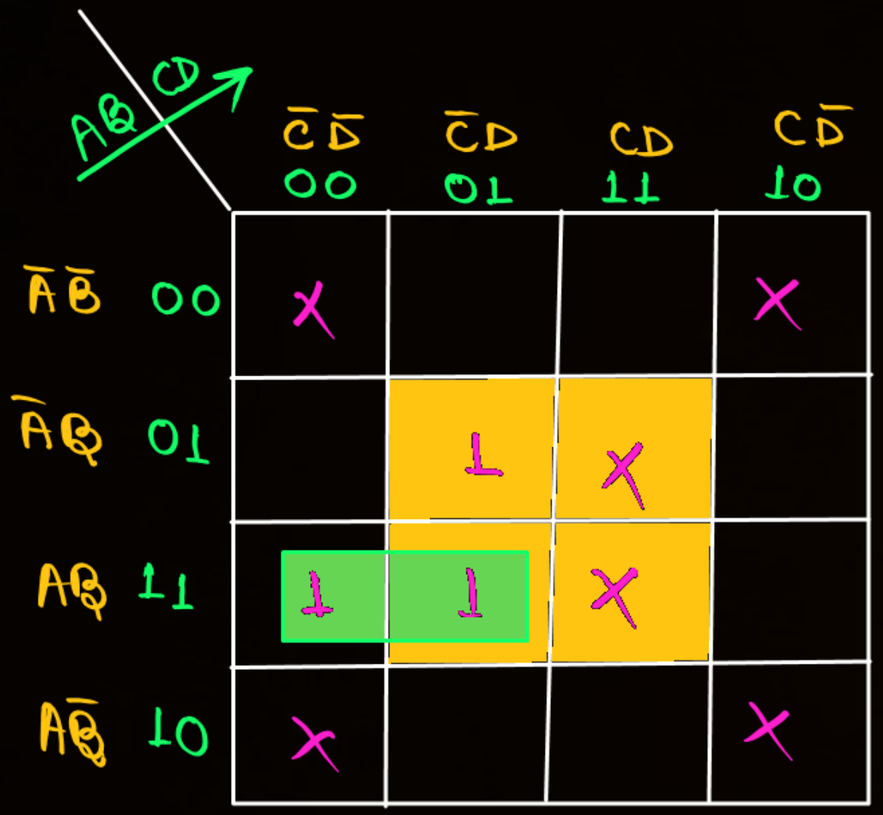


$$= \bar{A}\bar{D} + C\bar{D} + A\bar{B}C$$



$$= \bar{A}\bar{C} + C\bar{D} + A\bar{B}C$$

Q

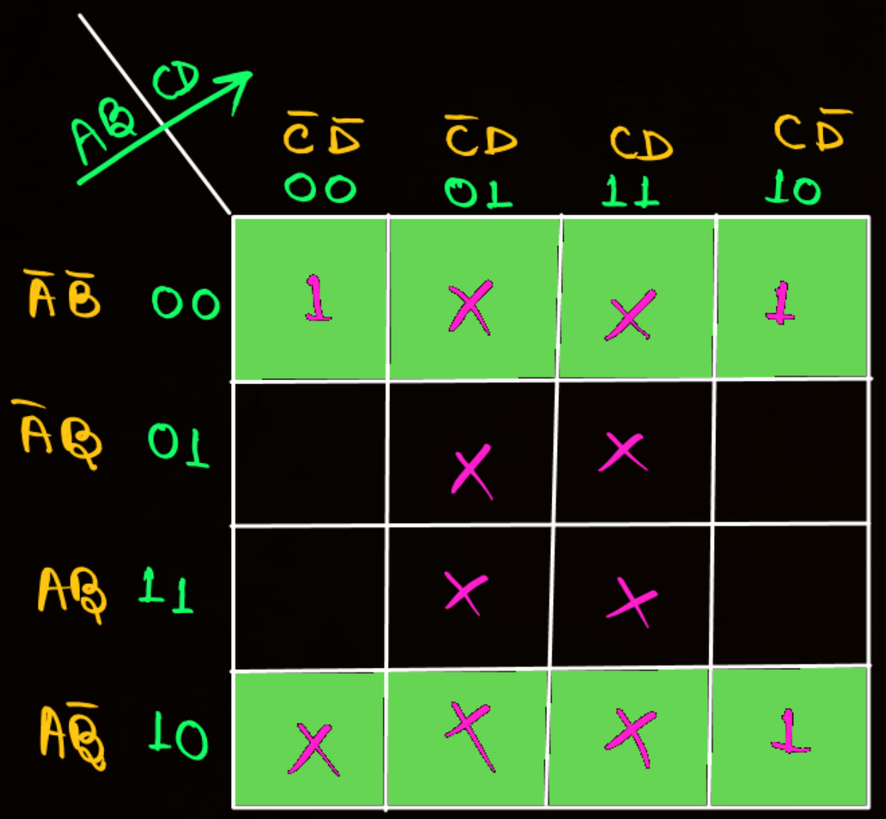


(A) One pair + 1 Quad

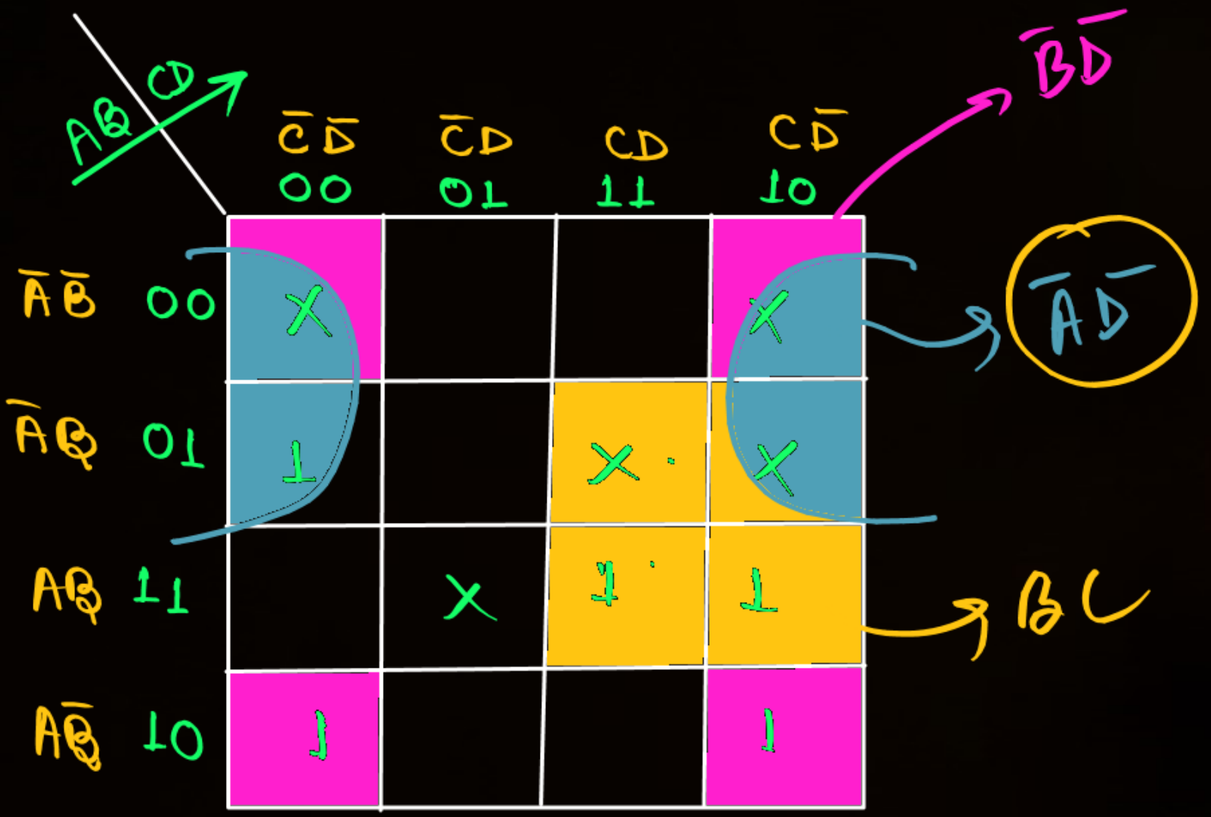
(B) One pair + 2 Quad

(C) 2 pair + 1 Quad

(D) None


AB
CD
$\bar{C}\bar{D}$ 00
$\bar{C}D$ 01
CD 11
$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00
$\bar{A}B$ 01
AB 11
$A\bar{B}$ 10
1
X
X
1
X
X
X
X
X
X
X
X
X
1

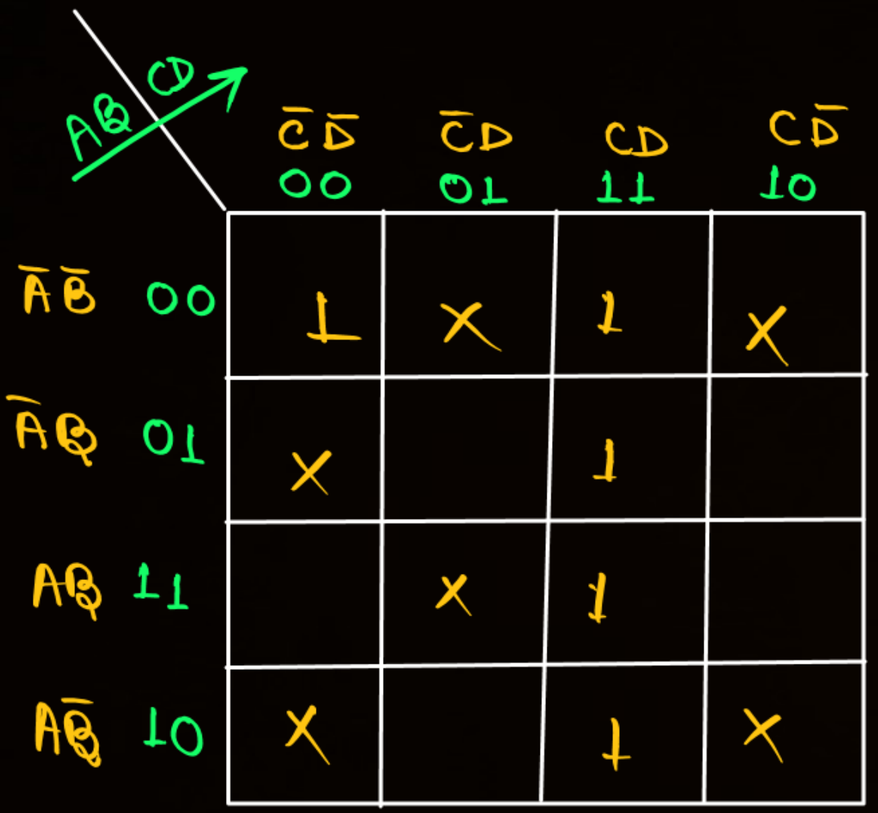
$= \bar{B}$



$$= BC + \bar{A}\bar{D} + \bar{B}\bar{D}$$

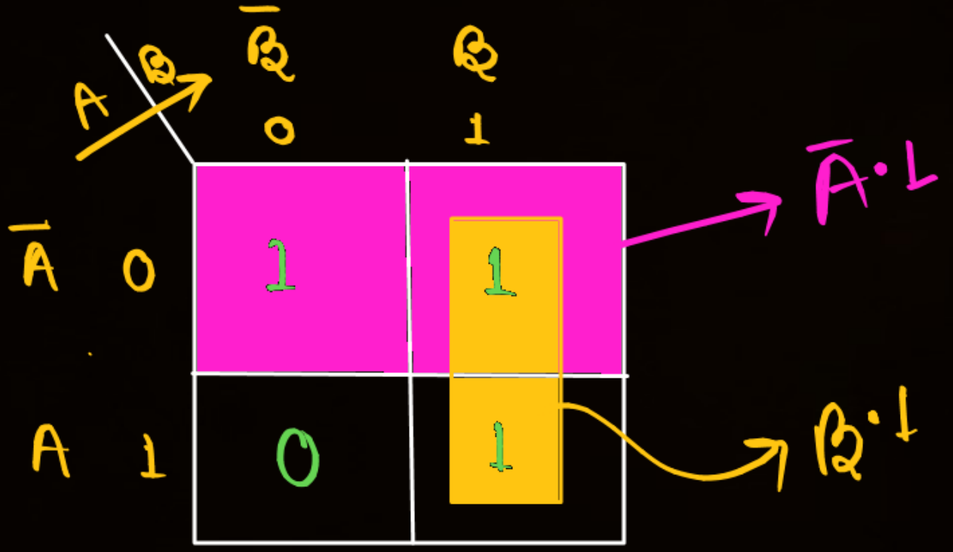
Ans

Q HW.


AB
CD
$\bar{C}\bar{D}$ 00
$\bar{C}D$ 01
CD 11
$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00
$\bar{A}B$ 01
AB 11
$A\bar{B}$ 10
1
X
1
X
X
1
1
X
X
1
1
X
X
1
X

$$f(A,B) = \bar{A}\bar{B} + \bar{A}B + A\bar{B} = \Sigma m(0,1,3)$$

$$= \bar{A}\bar{B}\cdot 1 + \bar{A}B\cdot 1 + A\bar{B}\cdot 0 + AB\cdot 1$$

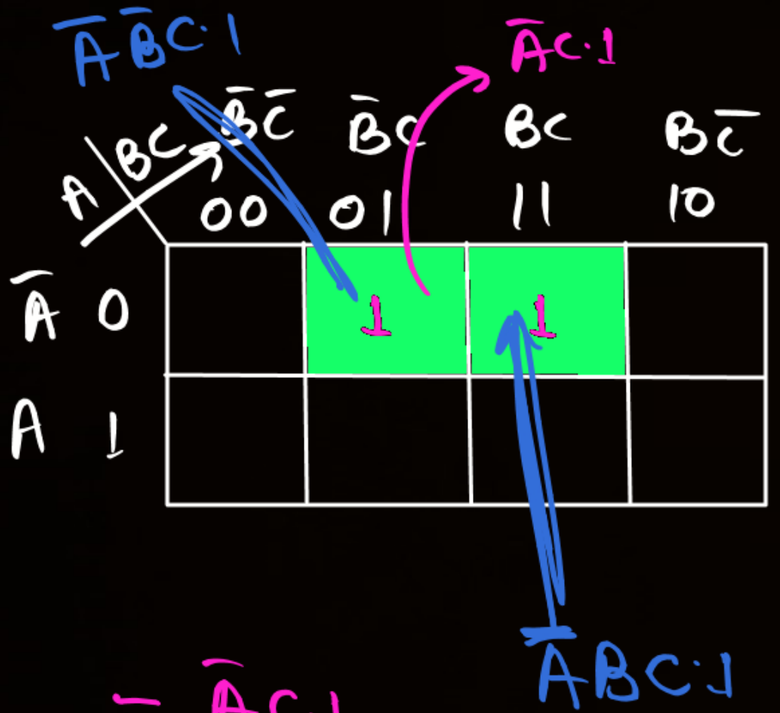


$\bar{A}\cdot 1$

$B\cdot 1$

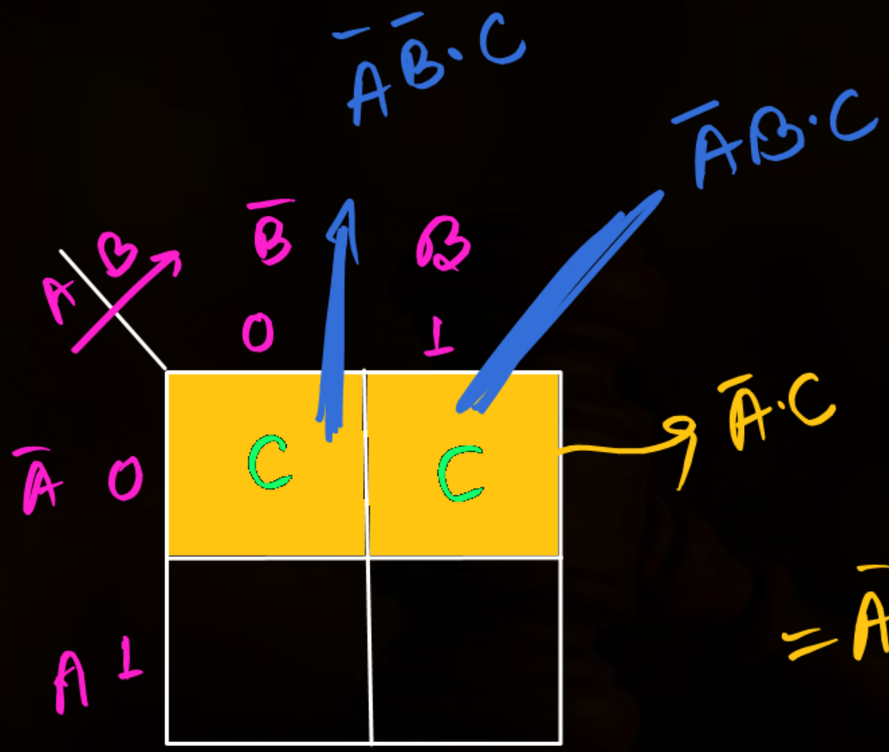
$$= \bar{A} + B$$

Q $f(A,B,C) = \bar{A}\bar{B}C + \bar{A}BC$



$= \bar{A}C:1$

$= \bar{A}C$

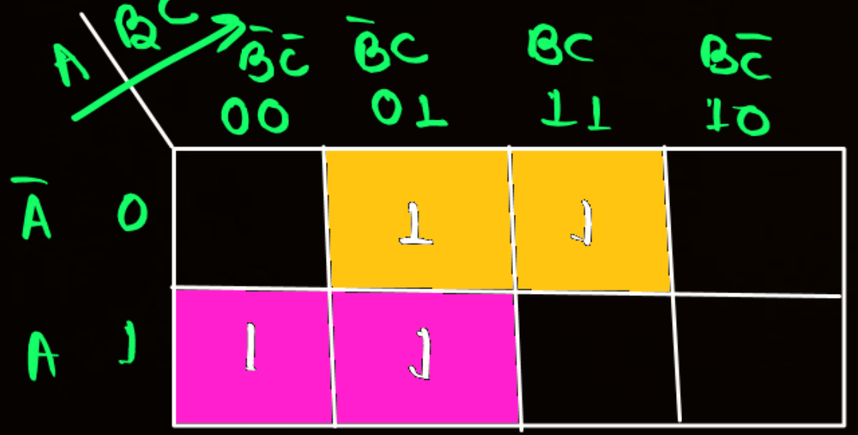


$= \bar{A}C$

Ex $f(A,B,C) = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}$

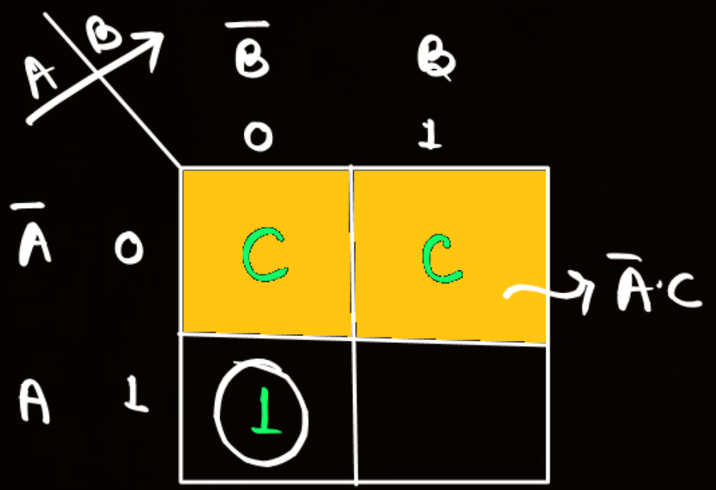
$= \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}(\bar{C}+C)$

$= \underset{001}{\bar{A}\bar{B}C} + \underset{011}{\bar{A}BC} + \underset{100}{A\bar{B}\bar{C}} + \underset{101}{A\bar{B}C} = \Sigma m(1,3,4,5)$



$= \bar{A}C + A\bar{B}$

Ex $f(A,B,C) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\cdot 1$

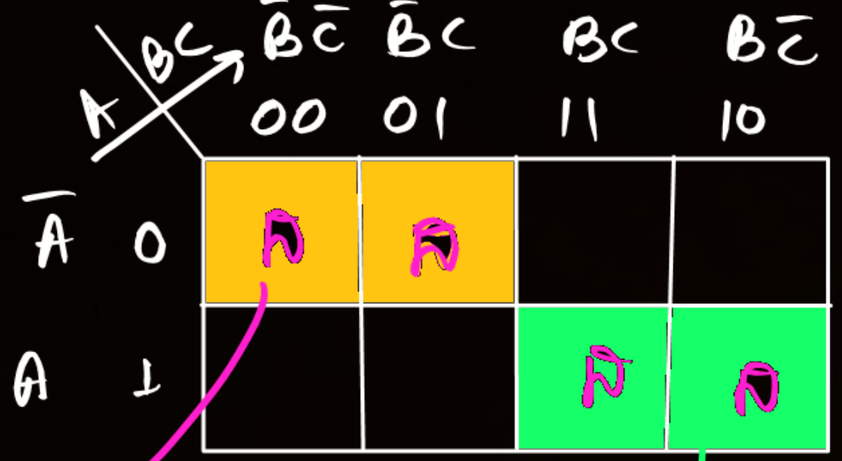


$A\bar{B}\cdot 1$

$= \bar{A}C + A\bar{B}$

$= \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + ABCD + AB\bar{C}D$

Q ≡



$= \bar{A}\bar{B}D + AB\bar{D}$

Answer



2 Minute Summary

PRIME IMPLICANTS

ESSENTIAL PRIME IMPLICANTS

Questions Practice

Discussion

Thank you

GW

Soldiers !

